

AhamX Bodhi

Where I Becomes Infinite

Spectrogram and Audio Features

Module 4.2: Audio and Speech Models

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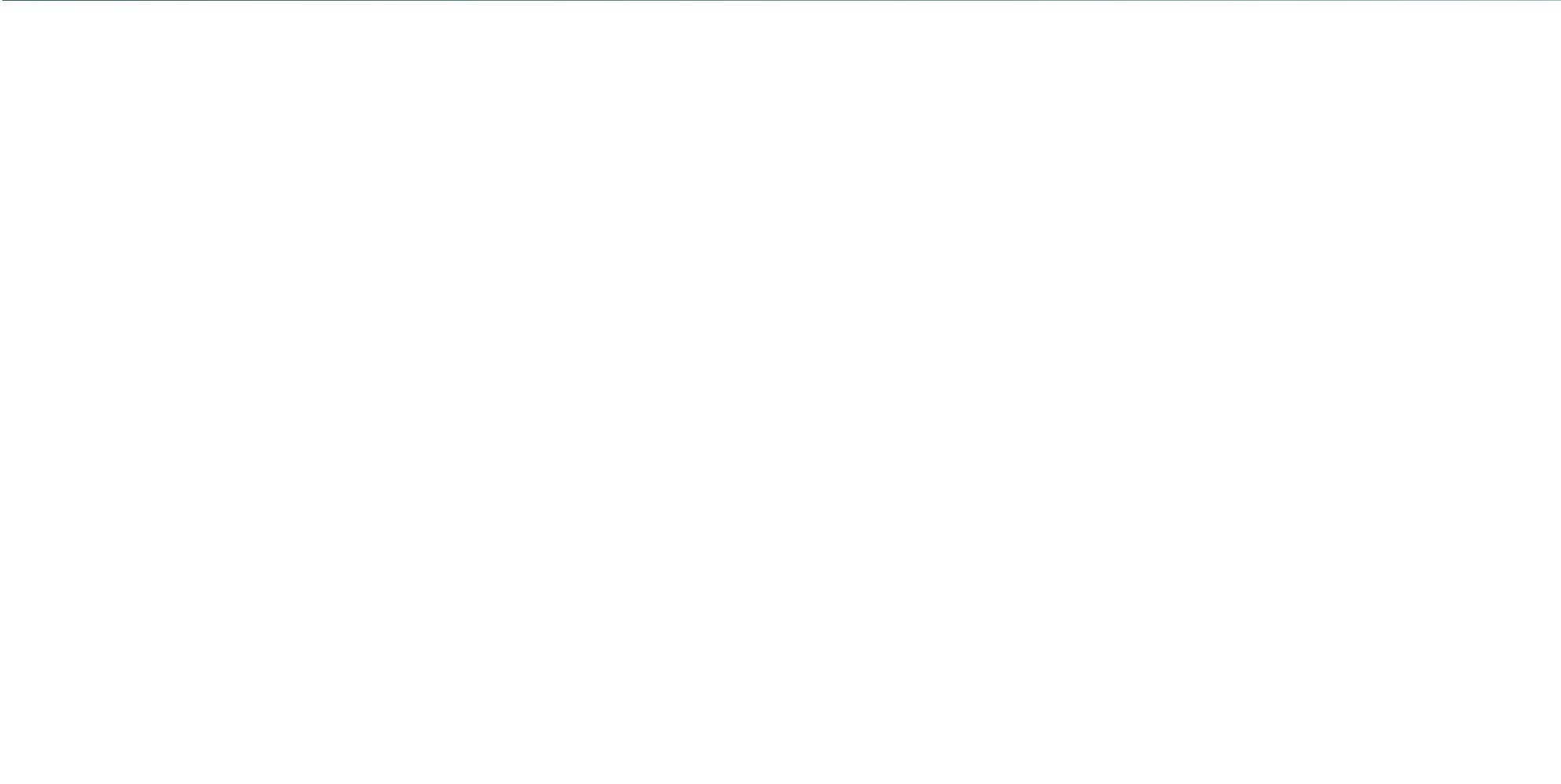
Session Overview

What You Will Learn

- Spectrogram variants: linear, mel, CQT, VQT
- Self-supervised audio features (wav2vec, HuBERT)
- How spectrograms are used as ViT inputs
- Learned vs. hand-crafted audio features
- Feature engineering for speech vs. music

Concept at a Glance

- **Duration:** 9 minutes
- **Bloom's Level:** B3 – Apply
- **Difficulty:** L2 – Intermediate
- **Module:** 4.2 Audio and Speech Models
- **Prerequisite:** Audio Processing Fundamentals (TH-000006)





Spectrogram Variants: A Comparison

Type	Frequency Axis	Best For	Used In
Linear	Uniform Hz spacing	General analysis	STFT visualization
Mel	Perceptual (mel scale)	Speech / ASR / TTS	Whisper, Tacotron
Log-frequency	Log Hz spacing	Music pitch	Music transcription
CQT	Constant-Q (geometric)	Musical chords	Music generation
Chromagram	12 pitch classes	Chord recognition	Music analysis

The mel-spectrogram is the dominant choice for speech AI. CQT/chromagram are preferred for music. Linear STFT is used as an intermediate step for reconstruction (Griffin-Lim, HiFi-GAN vocoder).



Mel-Spectrogram: Design Choices That Matter

Parameter Sensitivity for AI Engineers

- **n_mels (bins):** 80 bins (ASR standard), 128 bins (TTS/music quality). More bins = finer frequency resolution but larger feature matrix.
- **n_fft:** FFT window size. 512 (32ms at 16kHz) balances time and frequency resolution. Larger → better frequency resolution, worse temporal resolution.
- **hop_length:** Shift between windows. 160 samples (10ms) gives 100 frames/sec – standard for ASR. 256 (16ms) for TTS generation.
- **f_min / f_max:** Filter range. Default: 0 – Nyquist. For speech: f_min=80Hz, f_max=8000Hz to exclude irrelevant frequencies.
- **Normalization:** Per-utterance mean-variance normalization (CMVN) is critical for robustness to recording conditions and microphone differences.

Using the wrong hop_length is a common bug – check that model frame rate matches your feature extraction rate or you will mis-align audio and text timestamps.



Audio Spectrogram Transformer (AST): Audio as Image

Treating Spectrograms as Images for Transformer Processing

- **Key insight:** A mel-spectrogram is a 2D matrix (frequency $\times$ time) – exactly the same format as a grayscale image. ViT can process it without modification.
- **AST (Gong et al., 2021):** Fine-tuned ViT-B/16 on ImageNet-pretrained weights applied to mel-spectrograms. State-of-the-art on AudioSet classification.
- **Patch overlap:** AST uses overlapping patches (16×16 with stride 10) to increase spatial resolution – more patches = richer temporal and frequency coverage.
- **Cross-domain transfer:** ImageNet pre-training transfers surprisingly well to audio spectrograms – visual texture features in the network correspond to spectro-temporal patterns.
- **AudioMAE:** Self-supervised masked autoencoder pre-training on audio spectrograms. Achieves 86.6% mAP on AudioSet without ImageNet pre-training.

The spectrogram-as-image paradigm explains why audio AI models can be built by fine-tuning vision model architectures – dramatically reducing development time.





wav2vec 2.0: Self-Supervised Speech Representation

Learning from Raw Waveform (Baevski et al., 2020 – Meta AI)

- **Input:** Raw waveform (not spectrogram) fed into a CNN feature extractor producing 512-dim representations at 20ms frames
- **Self-supervised objective:** Contrastive task – predict which of K quantized speech units is the true next frame, given masked context. Similar to BERT's masked prediction but for audio.
- **Codebook:** Gumbel-softmax quantization produces discrete speech units ("audio tokens") from continuous features – enables training without transcripts
- **Fine-tuning:** After pre-training on 53k hours unlabeled audio, fine-tuning with only 10 minutes of labeled speech achieves 4.8% WER on LibriSpeech – remarkable data efficiency
- **Impact:** Enabled ASR for low-resource languages where collecting labeled speech is expensive. Used in Facebook's ASR for 100+ languages.

wav2vec 2.0 demonstrated that high-quality speech features can be learned from unlabeled audio – eliminating the need for expensive transcription for model pre-training.



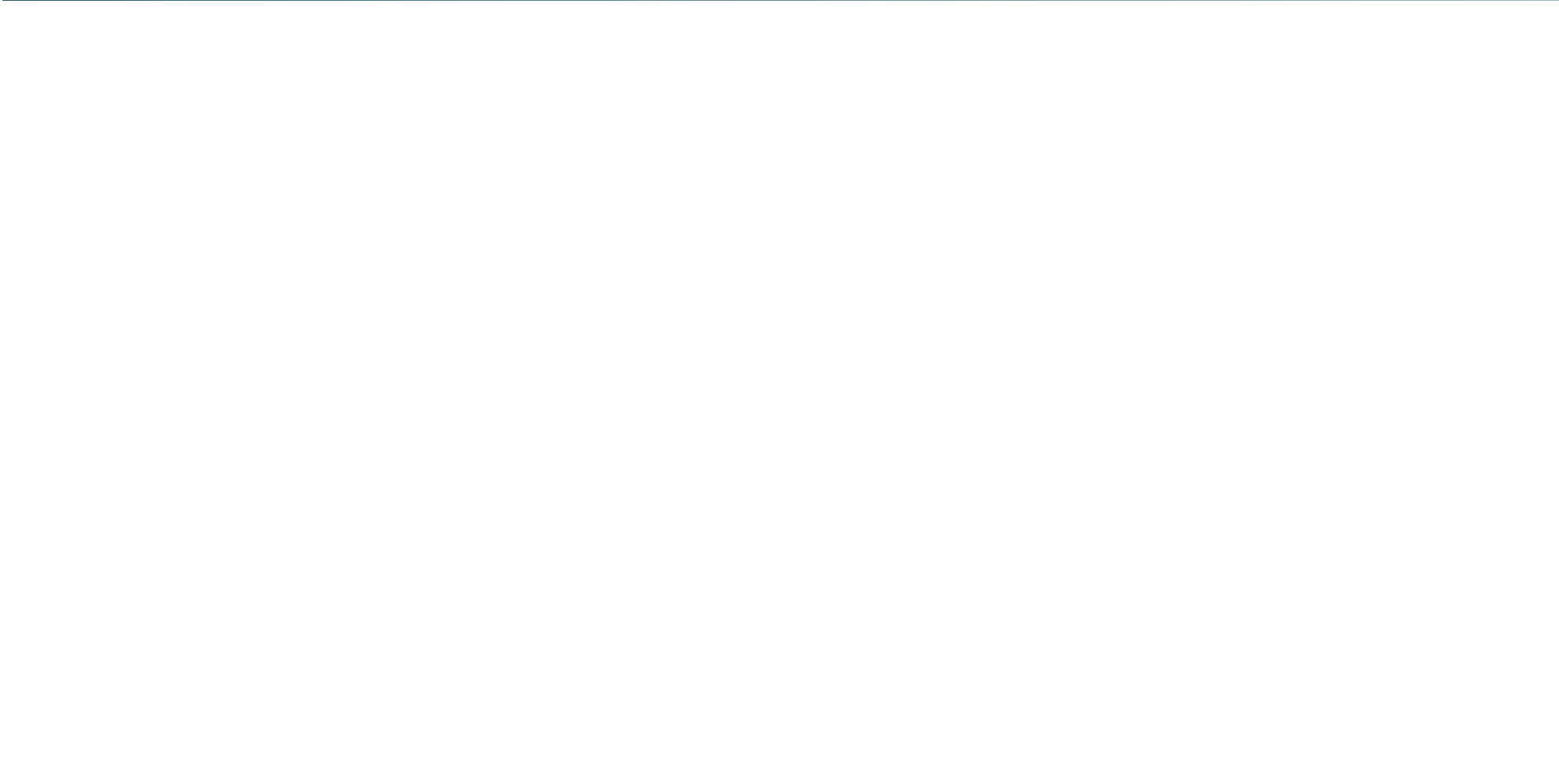
HuBERT: Hidden-Unit BERT for Speech

HuBERT (Hsu et al., 2021 – Meta AI)

- **Insight:** wav2vec 2.0 requires online quantization during training – complex and sensitive. HuBERT decouples feature learning from clustering.
- **Process:** (1) Cluster acoustic features (k-means on MFCCs/offline HuBERT features) to get pseudo-labels. (2) Train BERT-style masked prediction on pseudo-labels from raw waveform. Iterate.
- **Result:** HuBERT Large achieves 1.9% WER on LibriSpeech clean – state-of-the-art at the time, competitive with supervised models
- **Speech tokenization:** HuBERT discrete units are used as “speech tokens” in speech language models (SpeechGPT, AudioPaLM, VALL-E) – the audio equivalent of BPE tokens

Engineer's Application

HuBERT features are the standard input for speech emotion recognition, speaker verification, and voice conversion tasks – the “BERT of speech” for feature extraction.





Whisper's Audio Feature Pipeline

Whisper Preprocessing (OpenAI, 2022)

- **Step 1:** Resample audio to 16kHz, mono. Pad or truncate to 30 seconds.
- **Step 2:** Log mel-spectrogram with $n_{\text{mels}}=80$, $n_{\text{fft}}=400$ (25ms), $\text{hop}=160$ (10ms), $f_{\text{min}}=0$, $f_{\text{max}}=8000\text{Hz}$
- **Step 3:** Normalize: clip to $[-4.0, +4.0]$ log scale, scale to $[-1.0, +1.0]$
- **Step 4:** Output shape: $[80, 3000]$ – 80 mel bins, 3000 frames (30 sec at 100fps)
- **Encoder:** Two 1D Conv layers then standard ViT Transformer with sinusoidal positional embeddings – treating spectrogram “columns” as a 1D sequence
- **No language-specific preprocessing:** Whisper’s multilingual model handles 99 languages with the same mel-spectrogram input – language is predicted as a special token

Whisper’s preprocessing is open-source and battle-tested for production. Always use the official `whisper.load_audio()` and `whisper.log_mel_spectrogram()` functions for consistency.



GenAI Application: Environmental Sound Classification

Mel-Spectrogram + AST for Sound Event Detection

- **Task:** Classify audio clips into categories: dog bark, siren, gunshot, glass breaking – used in smart home devices, security systems, urban sound monitoring
- **Architecture:** Mel-spectrogram (128 bins) → Audio Spectrogram Transformer (AST-S) fine-tuned from ViT-S/16 ImageNet weights
- **Dataset:** AudioSet (Google, 527 classes, 2M clips) is the standard pre-training dataset. ESC-50 and UrbanSound8K for fine-tuning evaluation.
- **Results:** AST achieves 0.485 mAP on AudioSet – significantly better than CNN-based models. PaSST (2022) pushes this to 0.496 with patch shuffling augmentation.
- **Production deployment:** Run on-device (AST-Tiny, 5M params) for always-on sound detection with <10ms latency on edge hardware

Mel-spectrogram + ViT is now the go-to architecture for audio classification tasks, replacing CNN architectures that dominated until 2021.

GenAI Application: Neural Music Generation



Audio Features in Music Generation Systems

- **EnCodec (Meta, 2022):** Neural audio codec that compresses waveforms into discrete tokens at multiple bitrates. Encodes 24kHz audio at 1.5kbps with near-transparent quality. Used as the token vocabulary for music LMs.
- **MusicGen (Meta, 2023):** Autoregressive Transformer conditioned on text and optional melody. Uses EnCodec tokens as the “audio vocabulary” – music generation = token generation.
- **AudioCraft:** Open-source toolkit containing MusicGen, AudioGen (sound effects), and EnCodec. Engineers can fine-tune MusicGen on custom music styles.
- **Stable Audio (Stability AI, 2024):** Latent diffusion model on mel-spectrograms conditioned on text and timing metadata. Generates 45-second clips at 44.1kHz.

Key Insight

Neural codecs (EnCodec, DAC) have replaced spectrograms as the audio representation in generative models – discrete tokens enable autoregressive generation, not possible with continuous spectrograms.



GenAI Application: Voice Cloning with Speaker Embeddings

Zero-Shot Voice Cloning Architecture

- **Task:** Given 3–10 seconds of target speaker audio, synthesize speech in that speaker's voice from arbitrary text
- **Speaker embedding:** Extract mel-spectrogram features → speaker encoder (d-vector, x-vector, or ECAPA-TDNN) → 256-dim speaker embedding that captures voice timbre and accent
- **TTS conditioning:** Speaker embedding injected into TTS model (YourTTS, VITS, Tortoise) as conditioning vector at every decoder step
- **State of the art:** VALL-E (Microsoft, 2023) uses EnCodec tokens and a language model conditioned on 3-second speaker prompt – zero-shot, highly realistic
- **Ethical consideration:** Speaker embeddings can be used for voice deepfakes. Responsible deployment requires consent verification and watermarking (SynthID Audio from Google).

Understanding mel-spectrogram and speaker embedding extraction is essential for AI engineers building voice products with personalization features.



Key Takeaways

What You Should Remember

- Mel-spectrogram (80–128 mel bins, 10ms hop) is the standard input representation for ASR (Whisper) and TTS models
- Spectrograms can be treated as images and processed with ViT – Audio Spectrogram Transformer (AST) achieves state-of-the-art on AudioSet
- Self-supervised models (wav2vec 2.0, HuBERT) learn speech features from unlabeled audio, producing discrete speech units used as tokens in audio LMs
- Neural codecs (EnCodec, DAC) produce discrete audio tokens enabling autoregressive music and speech generation
- CMVN normalization, VAD preprocessing, and consistent hop_length are critical engineering details that separate prototype from production quality

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